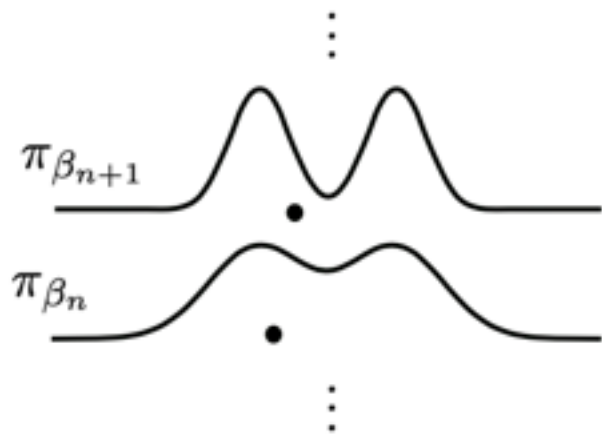
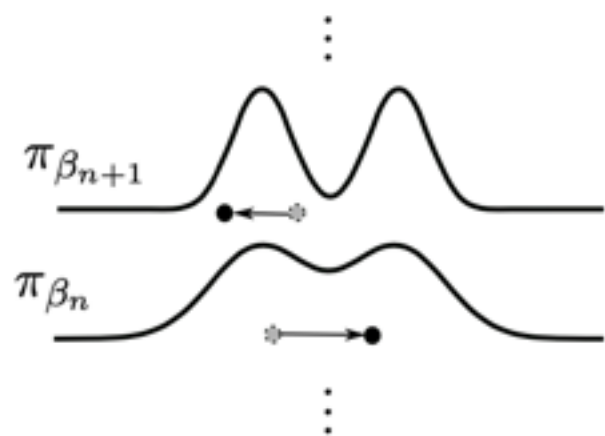


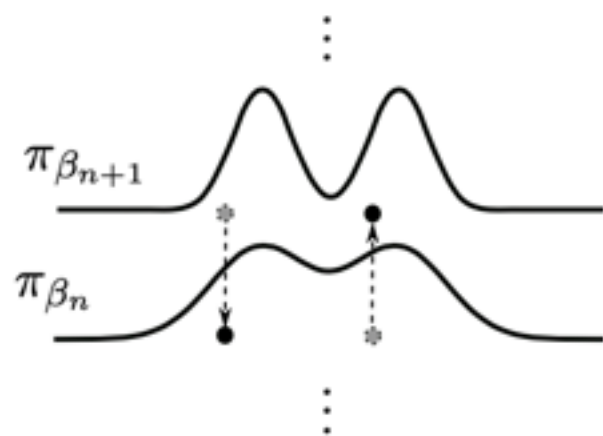
Initialize



Local Exploration



Communication



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- Accept/reject using Metropolis-Hastings:

$$\mathbf{K}^{(n,n+1)}(\mathbf{x}, d\mathbf{x}') = \alpha^{(n,n+1)}(\mathbf{x})\delta_{\mathbf{x}^{(n,n+1)}}(d\mathbf{x}') + (1 - \alpha^{(n,n+1)}(\mathbf{x}))\delta_{\mathbf{x}}(d\mathbf{x}')$$

- The n -th swap move takes the state $\mathbf{x}$ proposes $\mathbf{x}^{(n,n+1)}$ swapping components n and $n + 1$

$$\mathbf{x}^{(n,n+1)} = (x^0, \dots, x^{n-1}, x^{n+1}, x^n, x^{n+2}, \dots, x^N)$$

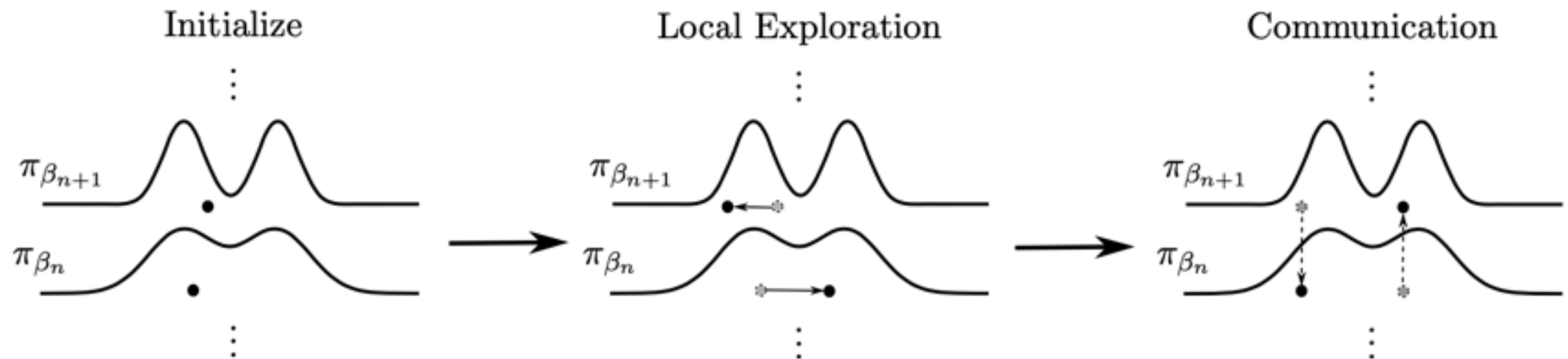


$$a^{(n,n+1)}(\mathbf{X}) = 1 \wedge \frac{\pi(\mathbf{X}^{(n,n+1)})}{\pi(\mathbf{X})}$$

$$= 1 \wedge \frac{\pi_{\beta_n}(x^{n+1})\pi_{\beta_{n+1}}(x^n)}{\pi_{\beta_n}(x^n)\pi_{\beta_{n+1}}(x^{n+1})}$$

► Acceptance probability only depends on likelihood:

SWAP MOVE



- The n -th swap move takes the state $\mathbf{x}$ proposes $\mathbf{x}^{(n,n+1)}$ swapping components n and $n + 1$

$$\mathbf{x}^{(n,n+1)} = (x^0, \dots, x^{n-1}, x^{n+1}, x^n, x^{n+2}, \dots, x^N)$$

- Accept/reject using Metropolis-Hastings:

$$\mathbf{K}^{(n,n+1)}(\mathbf{x}, d\mathbf{x}') = \alpha^{(n,n+1)}(\mathbf{x})\delta_{\mathbf{x}^{(n,n+1)}}(d\mathbf{x}') + (1 - \alpha^{(n,n+1)}(\mathbf{x}))\delta_{\mathbf{x}}(d\mathbf{x}')$$

- Acceptance probability only depends on likelihood:

$$\alpha^{(n,n+1)}(\mathbf{x}) = 1 \wedge \frac{\pi(\mathbf{x}^{(n,n+1)})}{\pi(\mathbf{x})} = 1 \wedge \frac{\pi_{\beta_n}(x^{n+1})\pi_{\beta_{n+1}}(x^n)}{\pi_{\beta_n}(x^n)\pi_{\beta_{n+1}}(x^{n+1})}$$

COMMUNICATION